

SMS demo project spec

This is the capstone for the Epic Learn **Go Programming Master Course**. The codebase is this repository ([160221J/sms](#)). The full week-by-week plan is in [COURSE.md](#) .

Product

Student Management System for campus staff:

- Staff **register** and **login** (JWT).
- Authenticated staff **CRUD students** (name, email, phone, course, date of birth).
- Simple React client for login/register/dashboard. Student UI is a stretch.

Architecture (target)

```
browser / Postman
  → Gin (:8080)
    → middleware (CORS, JWT)
    → handlers
    → repository
    → PostgreSQL (sms)
```

Package	Role
<code>cmd/server</code>	Process entry, CORS, listen
<code>internal/config</code>	Env: DB, JWT, port
<code>internal/database</code>	<code>pgx</code> pool
<code>internal/routes</code>	<code>/health</code> , <code>/api/register</code> , <code>/api/login</code> , <code>/api/students</code>
<code>internal/handlers</code>	HTTP + validation
<code>internal/repository</code>	SQL
<code>internal/models</code>	DTOs and entities
<code>internal/middleware</code>	Bearer JWT
<code>internal/utils</code>	JWT sign/verify
<code>frontend/</code>	Vite React client (not the grading core)

Classroom build order

Git history built auth early. **Class order** (so week 5 has a demo without functions/SQL mastery):

1. Weeks 4–5: module layout, `GET /health`
2. Weeks 6–8: functions, student structs, in-memory CRUD, repository
3. Week 9: errors, `slog`, config
4. Weeks 10–12: races, mutex, context; optional bulk import
5. Week 13: tests + CI
6. Week 14: PostgreSQL + migrations
7. Week 15: JWT, protect routes, run `frontend/`
8. Week 16: review and demo

Definition of done (week 16)

- ☐ Register, login, student CRUD against PostgreSQL
- ☐ JWT on student routes
- ☐ Parameterized SQL; schema in `backend/migrations/`
- ☐ `.env` gitignored; `.env.example` present
- ☐ `gofmt + go test ./...`
- ☐ README with run steps and example `curl`
- ☐ Oral walkthrough of one request

Known WIP gaps (assign these)

- `frontend/src/pages/Students.jsx` is a stub
- `internal/handlers/dashboard_handler.go` is empty
- No `*_test.go` files
- Logging is still `log` / Gin default, not `slog`
- No GitHub Actions yet